

Modernizing Test Automation

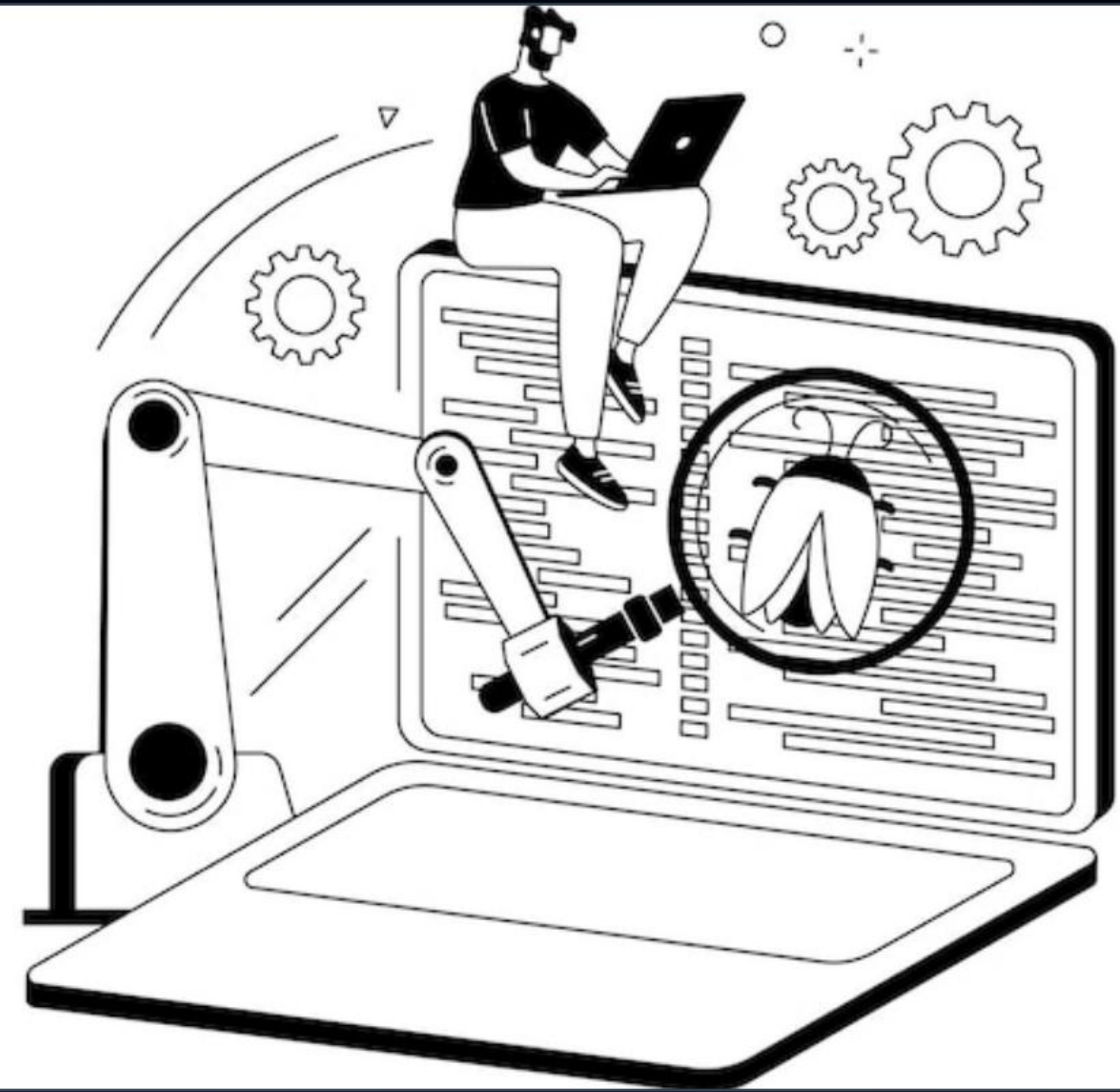
Capstone Project: Transitioning to AI-Assisted Open Source Frameworks

College Capstone Presentation | Salesforce CRM Team | Wells Fargo Internship

| PROBLEM & OBJECTIVE

Articulation (5 Marks): Legacy proprietary tools introduced high operational overhead and restricted developer-centric customizability within the Salesforce environment.

Objective: To implement a scalable, code-driven framework utilizing Playwright and AI to improve reliability and reduce dependency on external platform subscriptions.



TECHNOLOGY COMPARISON

Feature	Legacy Tool (ACCELQ)	Modern Framework (Playwright)
Approach	No-Code / Proprietary	Code-Driven (TypeScript/JS)
Performance	Limited by Cloud Infrastructure	Native Browser Contexts (High Speed)
Ecosystem	Vendor-Locked	Open-Source / Extensive Plugins
Debugging	Platform UI dependent	Advanced Trace Viewer & Snapshots
Cost Model	High Recurring Subscription	Optimized Internal Infrastructure

CORE TECHNOLOGY STACK



Playwright

Cross-browser automation engine supporting Chromium, WebKit, and Firefox for end-to-end CRM validation.



Codegen

Leveraged for rapid test scaffolding and element discovery, accelerating the migration of manual test cases.

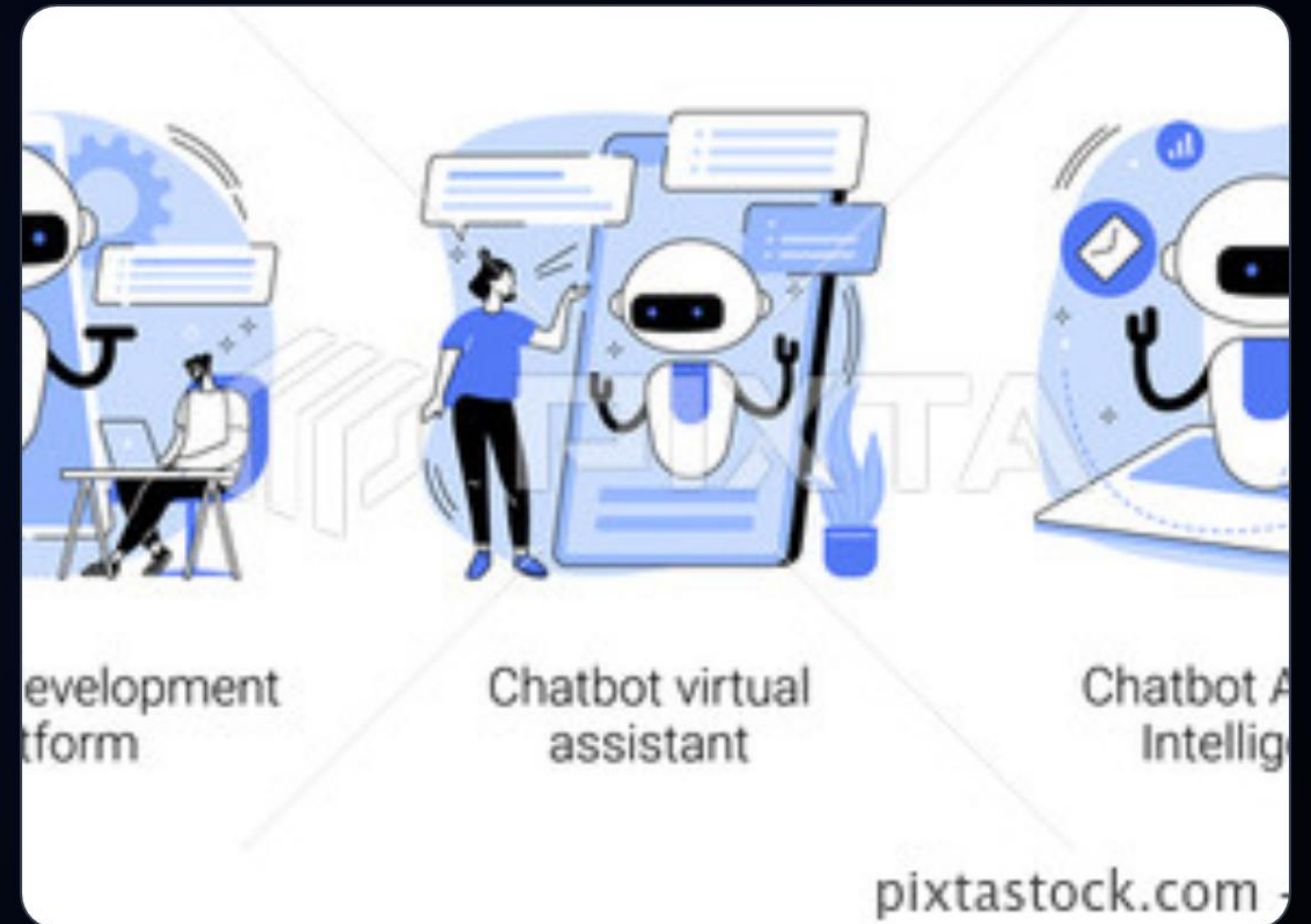


TypeScript

Implementing type-safe scripts to ensure maintainability and robustness across complex Salesforce DOMs.

| AI-ASSISTED DEVELOPMENT

- 🤖 **Playwright MCP:** Integrating Model Context Protocol to bridge LLMs with real-time browser states.
- 🔧 **GitHub Copilot:** Extensive utilization for logic refactoring and generating resilient locators.
- 🧠 **Agentic Testing:** Moving toward self-healing scripts that adapt to UI changes in the CRM platform.



| APPLICATION OF SCIENTIFIC PRINCIPLES

-  **Separation of Concerns:** Implementation of the Page Object Model (POM) to isolate UI elements from functional test logic.
-  **Asynchrony & Determinism:** Managing non-deterministic UI behavior via event-driven auto-waiting and retry-ability logic.
-  **Semantic Locators:** Using accessibility-first strategies (Role, Label) to build tests that mirror actual user interactions.
-  **Test Isolation:** Ensuring every execution starts with a clean browser context to prevent data leakage between test runs.

Quality & Quantum of Work

Migration of extensive CRM regression suites and development of a modular, reusable automation library for enterprise scalability.

ENGAGEMENT & WORK ETHICS

 Agile Team Board

Agile Collaboration

Active participation in Sprints, Stand-ups, and retrospectives within the Salesforce CRM team.



Professional Ethics

Adhering to enterprise security protocols and data privacy standards throughout the project lifecycle.

 Playwright Icon

Peer Reviews

Engaging in rigorous code reviews to maintain high-quality, standardized automation scripts.

SELF-EVALUATION & LEARNING

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"Transitioning from codeless platforms to a developer-centric AI framework has refined my understanding of software engineering trade-offs and the future of agentic automation."

— Intern Reflection

Learning outcomes: Advanced TypeScript, AI prompt engineering, and framework design patterns.

| PROJECT VALUE & IMPACT

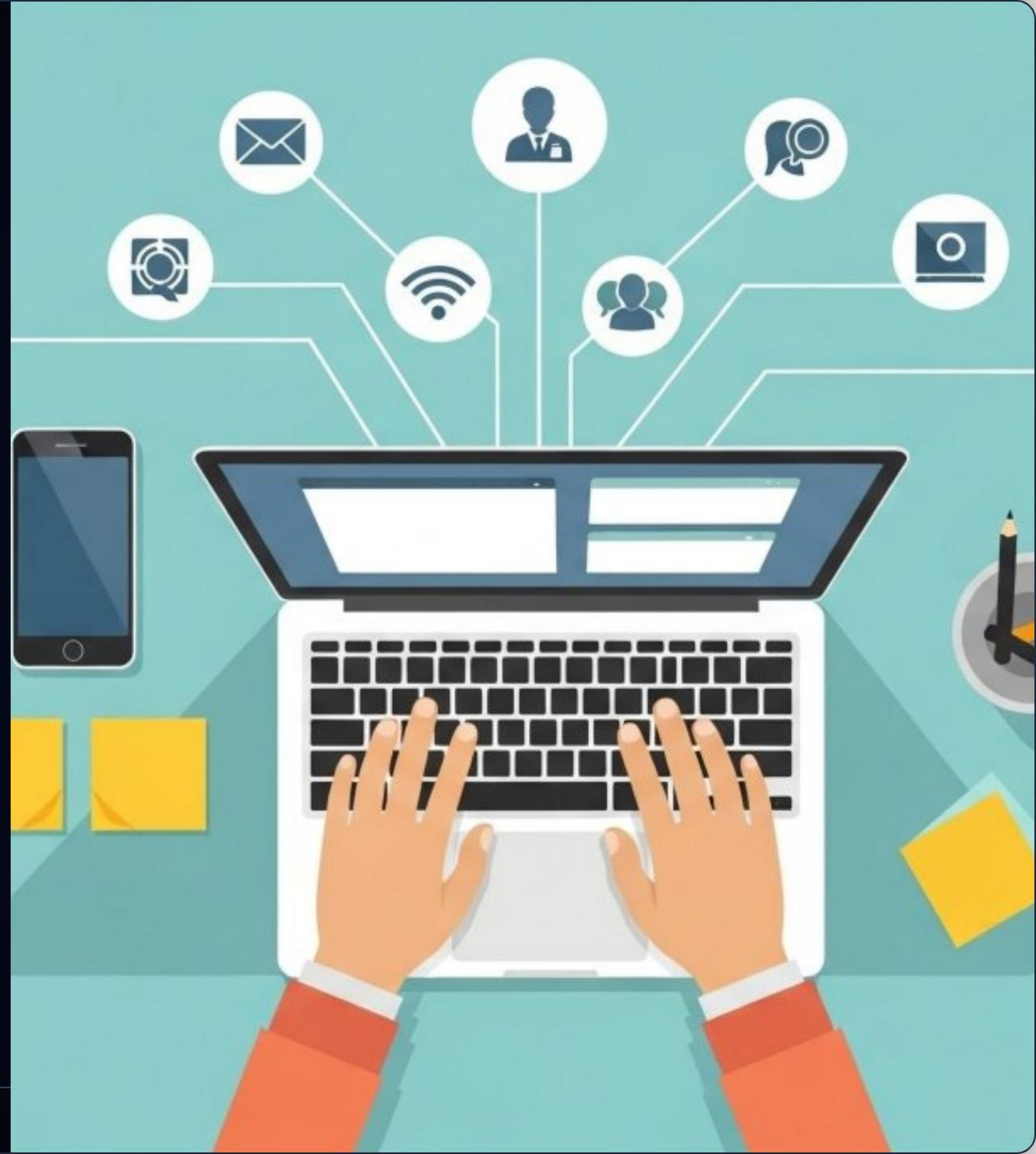
100%
Open Source Alignment

Key Accomplishments

- ✓ Elimination of proprietary subscription overhead.
- ✓ Significant reduction in test execution time.
- ✓ Improved framework flexibility for Salesforce updates.
- ✓ Seamless integration into the enterprise DevOps pipeline.

TECHNICAL WORKFLOW

- 1 Prototyping:** Capture browser events via Playwright Codegen.
- 2 Logic Refactoring:** Convert skeletons into modular Page Objects.
- 3 AI Optimization:** Refine locators using Playwright MCP and Copilot.
- 4 Continuous Delivery:** Triggering automated suites upon CRM releases.



Questions?

Thank you for your time and feedback.

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IMAGE SOURCES



https://img.freepik.com/free-vector/automation-testing-abstract-concept-vector-illustration-software-testing-solution-development-process-automation-website-developer-optimization-online-service-site-menu-bar-abstract-metaphor_335657-6140.jpg

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